

WORK SCHEDULE

Companion Guide

Understanding the Terms, Concepts, and Strategy

This document explains every metric, symbol, and design principle on the one-page Work Schedule. It is written for anyone who needs to understand the strategy — whether you are technically minded or not. Each concept is first explained in plain language with a real-life analogy, then supported with the precise definition and numbers.

All numerical values are produced by *cnhr_calculator.py*, the reference Python implementation that serves as the single source of truth for the framework.

1. The Core Idea

The CNHR-MNHR Framework is a mathematical system for answering one question: how many days should a private hire vehicle (PHV) driver work each week to meet legal minimum wage requirements while preserving as much free time as possible?

The framework exists because this question is harder than it looks. A PHV driver faces two types of cost: costs that run every day regardless of whether the car moves (insurance, vehicle finance, licensing), and costs that only arise when driving (electricity to charge the car, tyre wear, congestion charges). The first type — fixed costs — creates a mathematical trap.

The Rent Analogy

Imagine you rent a market stall for £220 per week. The rent is the same whether you open for 1 day or 7 days. If you open for just 2 days, each day must cover £110 of rent before you even start making profit. If you open for 5 days, each day only needs to cover £44 of rent. Opening more days makes each day cheaper — not because you sell more per day, but because the fixed rent is spread more thinly.

PHV driving works exactly the same way. The 'rent' is £220 per week in insurance, vehicle finance, and other fixed costs. Each working day must pay its share of this rent before generating any real income. The framework calculates exactly how many days you need to open the stall to cover the rent and meet the legal minimum wage — and not one day more.

Current configuration: 4 days on (Monday–Thursday), 3 days off (Friday–Sunday), every week identical. 10-hour shifts from 04:00 to 14:00. This produces 40 clock hours per week, targeting £203 per working day.

The underlying principle: higher earning rates do not increase your hourly income under this strategy — they reduce the number of days you need to work. Every productivity gain is converted into free time, not higher pay. The pay stays fixed at the legal minimum. This is a deliberate design choice made to maximise time available for study and career transition.

2. Why a 3-Week Cycle?

The schedule repeats every 3 weeks (21 days), not every 1 or 2 weeks. This is not arbitrary — it exists because of the fallback pattern.

The primary plan is to work 12 days per cycle (4 per week, every week identical). But what if earnings fall below the target? The framework includes a fallback that increases to 13 working days. The number 13 does not divide evenly into weekly blocks: $13 \div 7 = 1$ remainder 6. You cannot split 13 days evenly across identical weeks. The smallest cycle that can accommodate both 12 days (**4 + 4 + 4**) and 13 days (**5 + 5 + 3**) as whole numbers is 21 days — three weeks.

The Calendar Puzzle

Suppose you have 13 marbles and 3 jars (one per week). You want to distribute the marbles so that no jar has too many (to avoid exhaustion) and the rest days fall on specific days. The best split is 5–5–3: two heavier weeks followed by a lighter recovery week.

If the cycle were only 7 days (1 week), you could not fit 13 days at all. If it were 14 days (2 weeks), 13 out of 14 would leave only 1 rest day — unsustainable. 21 days is the minimum that gives both the primary (12-day) and fallback (13-day) patterns enough room to breathe.

In normal operation ($d=12$), every week looks the same: Monday–Thursday on, Friday–Sunday off. The 3-week structure is invisible — it only becomes apparent if the fallback is activated. Think of it as a contingency built into the calendar.

3. The Weekly Pattern

Primary pattern ($d=12$): Work Monday through Thursday, rest Friday through Sunday. Every week identical.

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Week 1	WORK	WORK	WORK	WORK	REST	REST	REST
Week 2	WORK	WORK	WORK	WORK	REST	REST	REST

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Week 3	WORK	WORK	WORK	WORK	REST	REST	REST

Working: 12 days (4+4+4). Rest: 9 days (3+3+3). Maximum consecutive working days: 4 (Monday–Thursday). 40 clock hours per week. 8.66 engaged hours per shift.

Why these days? Each day of the week was scored 0–20 based on real demand drivers: commuter patterns, airport schedules, school runs, and traffic flow. Monday and Tuesday scored highest (14/20); Wednesday and Thursday are strong (11–12). Friday, Saturday, and Sunday scored lowest (6–10). The working days are the highest-demand days and the rest days are the lowest — maximising earnings per shift while protecting the weakest days for rest.

4. A Working Day

Every working day follows the same structure:

Time	Block	Duration	What Happens
03:45	Wake	15 min	Prepare for shift
04:00–14:00	SHIFT	10 hours	Active driving. Uber app on. Earning, waiting for trips, or repositioning.
14:00–15:00	Transition	1 hour	Drive home, park, wind down. Not earning but not yet free.
15:00–20:00	Free time	5 hours	Uncommitted waking hours. Study, rest, pray, cook, errands — your choice.
20:00–04:00	Sleep	8 hours	Non-negotiable. Circadian-aligned rest.

On rest days, you have 16 hours of free time (24 hours minus 8 hours of sleep). The shift and transition blocks disappear entirely.

Where Does the Time Go?

In a typical working week (Mon–Thu on, Fri–Sun off), across 7 days:

Sleep: 56 hours (8h × 7 days) — same every week

Work + transition: 44 hours (11h × 4 days) — the cost of earning

Free time: 68 hours (5h × 4 work days + 16h × 3 rest days)

Out of 168 hours in the week, 68 are genuinely yours. That is 9.71 hours per day on average — though no single day delivers exactly that. Working days give 5; rest days give 16.

5. The Parameters Explained

The framework uses a small set of measured numbers to calculate everything. Each was derived from 735 actual trips across 13 weeks of driving.

5.1 Trip Earning Rate — ρ (rho)

ρ = £29.55 per hour. The average fare earned per hour of active trip time — the clock runs only when you have a passenger or are driving to pick one up.

What ρ Actually Measures

If you spend exactly 1 hour with passengers (or en route), you earn £29.55 on average. But you do not spend your entire shift with passengers. Much of the time you are waiting for the next request or repositioning. ρ only counts the productive minutes.

Think of it like a taxi meter: ρ is the average reading when the meter is running. It says nothing about how often the meter is off.

5.2 Engagement Multiplier — α (alpha)

α = 1.26. For every hour of trip time, 1.26 hours of total engaged time are required. The extra 0.26 hours covers waiting between trips, repositioning, queuing.

The Waiter Analogy

A restaurant waiter who serves tables for 6 hours might be at the restaurant for 7.5 hours — including setup, side work, and waiting. Their α would be $7.5/6 = 1.25$. They are working the whole time but only earning tips for 6 hours.

For PHV driving, $\alpha = 1.26$ means about 20% of your engaged time is overhead. This is remarkably efficient — Uber keeps you busy most of the time.

5.3 Trip Ratio — r

r = 0.687. Of a 10-hour shift, 68.7% is actual trip time (6.87 hours). The remaining 31.3% is on-shift but not on a trip — breaks, charging, dead periods, driving between zones.

How they connect: Daily gross = $\rho \times r \times h = £29.55 \times 0.687 \times 10 = £203$. The composite $k = r \times \alpha = 0.866$ converts 10 clock hours into 8.66 engaged hours — the number used for minimum wage calculations.

5.4 Fixed Cost — F

F = £31.43/day (£220/week). Runs every calendar day. Insurance, vehicle finance, phone, MOT, PCO licence, incidentals. On rest days, holidays, sick days — it never stops.

5.5 Variable Cost — v

v = £42.00 per working day. Incurred only when driving. Primarily charging cost (£8.83 per charge at off-peak rates). One 10-hour shift $\approx$ one full charge $\approx$ 200+ miles. Also covers congestion, tyres, maintenance, cleaning.

Fixed vs Variable: The Phone Bill Analogy

Your phone contract costs £30/month whether you make calls or not (fixed). Each call costs 5p per minute (variable). If you make no calls, you still pay £30. If you make lots of calls, you pay £30 plus the per-minute charges.

For PHV driving: insurance and vehicle rent are the £30 monthly fee. Charging the car is the per-minute call cost. Working more days does not increase the insurance bill — it just adds more charging cost. This is why the number of working days matters so much: more days spread the 'phone contract' across more shifts.

5.6 Working Density — δ (delta)

$\delta = 0.571$. The fraction of days that are working days. $12/21 = 0.571$. This is the single control variable. Once set, every financial and temporal metric follows. Think of it as the one dial you can turn.

6. What the Metrics Mean

6.1 CNHR — Your True Hourly Rate

CNHR(e) = £12.25/hr. After deducting every cost, this is what you earn per engaged hour.

The Payslip Number

If you were an employee, CNHR(e) would be the number on your payslip after all deductions. Not your gross fare (£29.55/hr), not your daily takings (£203), but the actual net rate once every cost of operating the vehicle has been subtracted.

At £12.25/hr, you are £0.04 above the National Living Wage. That razor-thin margin is deliberate — the framework trades income headroom for free time.

The formula has three terms:

Term	Value	What It Represents
ρ/α	£23.45/hr	The attractor — the maximum CNHR could reach with zero costs. Set by the market.
$v/(kh)$	£4.85/hr	Variable cost drag — charging and consumables per engaged hour. Same regardless of days worked.
$F/(kh\delta)$	£6.35/hr	Fixed cost drag — the £220 weekly burden spread across engaged hours. Shrinks as you work more days.

Why More Days = Higher Rate

The £220 weekly fixed cost must be paid regardless. At 4 working days (34.6 engaged hours), each hour carries £6.35 of this burden. At 5 working days (43.3 engaged hours), each hour carries only £5.08.

It is like splitting a restaurant bill: the same £220 split 4 ways = £55 each. Split 5 ways = £44 each. The bill has not changed; the share per person has.

6.2 MNHR — Trip-by-Trip Performance

MNHR (Marginal Net Hourly Rate) is the earning rate of each individual trip. It fluctuates wildly — a 5-minute minimum-fare ride might show £48/hr; a 90-minute airport run might show £18/hr. CNHR smooths this out by weighting each trip by its duration, avoiding a statistical bias called the mean-of-ratios problem.

6.3 NLW — The Legal Floor

NLW = £12.21/hr. The National Living Wage. Since the Supreme Court classified Uber drivers as workers, CNHR(e) must meet or exceed this. At the current configuration, headroom is £0.04/hr.

The Speed Limit

NLW is like a speed limit. You must not fall below it, but there is no reward for being far above it (under this strategy). Driving at exactly 70mph gets you there just as legally as 90mph. The framework aims for 70.04 — just barely legal, least fuel burned.

7. Time-Value Metrics

The financial metrics confirm the schedule is legal. The time-value metrics tell you whether it is worth living inside.

7.1 $H(\delta)$ — Free Hours Per Day

$H(\delta) = 9.71$ hours/day (average). Weighted average across working days (5h) and rest days (16h).
Formula: $H = 16 - 11\delta$. In a 7-day week: $4 \times 5 + 3 \times 16 = 68$ free hours = 9.71/day.

7.2 τ (tau) — The Exchange Rate

$\tau = 1.55$. For every clock hour committed to work (including transition), you receive 1.55 hours of free time. $\tau > 1$ means more free time than work time across the week.

The Currency Exchange

Think of τ as a bureau de change. Hand over 1 hour of work, receive 1.55 hours of freedom. That is a favourable rate.

Across a working week: you commit 44 hours ($4 \times 11h$) and receive 68 hours of freedom. At 5 working days, τ drops to 1.05. At 6 days, $\tau = 0.70$ — work-dominant.

7.3 π (pi) — The Price of Freedom

π answers: what income do you forgo for each free hour gained by not working an additional day?

Marginal $\pi = £14.64/hr$ (constant, the decision metric). Each rest day costs £161 of net income (gross minus variable cost) and provides 11 extra free hours. $£161/11 = £14.64$. Same whether it is your 9th or 15th working day — no diminishing returns.

Average $\pi = £6.24/hr$ (varies, descriptive only). Total net earnings $\div$ total free hours. Lower than marginal because it includes cost recovery overhead. At current δ , about 57% of each working day is just paying rent.

Should You Work an Extra Day?

Decision rule: if a free hour is worth more than £14.64, do not add a working day.

Career example: Cybersecurity traineeship study hour = $£35,000 \div 1,311 = £26.70$. Since $£26.70 > £14.64$, every study hour earns more than driving. Minimise working days.

Leisure example: If you value relaxation at £10/hr, then $£10 < £14.64$ — work the extra day. But NLW already forces 12 days regardless.

7.4 η (eta) — Study Conversion

$\eta \geq 0.55$. The fraction of free hours that must become productive study to meet 1,311 hours by November 2026. On working days: ~2.75h study out of 5h free. On rest days: ~8.8h out of 16h. The rest-day figure is demanding. η is tracked during the trial, not assumed.

8. The Decision Rule

Work if marginal π (£14.64) < value of a free hour. Otherwise, stop.

Since marginal π is constant, the decision is binary. Two forces resolve it:

NLW constraint: At $p = £29.55$, minimum δ for NLW = 11.93 days $\rightarrow$ rounds to 12. You cannot legally work fewer.

Career transition value: Study hour = £26.70 > £14.64. Every free hour spent studying earns more than driving. Minimise days; NLW sets floor at 12.

Why Not Just Work More and Save?

Working 15 days instead of 12 generates an extra £635/month. But it costs 33 hours of free time per cycle — worth $33 \times £26.70 = £881$ in study value. You gain £635 cash but lose £881 in future career value. The study trade is nearly 40% more valuable.

9. The Monitoring System

The schedule includes a built-in feedback system — not ad hoc rules but a mathematically grounded control system, like a thermostat.

9.1 The Thermostat Analogy

How a Thermostat Works

A thermostat has three parts: a target temperature (20°C), a sensor (thermometer), and an actuator (boiler). Sensor reads below target $\rightarrow$ boiler on. Reaches target $\rightarrow$ boiler off. Self-correcting.

The work schedule operates identically:

Target: NLW = £12.21/hr (the temperature you must maintain)

Sensor: Day-3 checkpoint at £608 cumulative (the thermometer)

Actuator: Add a working day / sacrifice a rest day (the boiler)

Below target on Wednesday evening $\rightarrow$ actuator fires (Friday becomes a working day). On target or above $\rightarrow$ rest days protected.

9.2 Property 1: The NLW Attractor

When you minimise work to the NLW boundary, CNHR(e) collapses to exactly £12.21 regardless of trip rate p . This is algebraic — substituting minimum δ back into the formula causes p to cancel. Higher productivity $\rightarrow$ fewer required days, not higher pay. The framework is a subsistence optimiser.

9.3 Property 2: Why Day 3 (Wednesday)?

Enough data: After 30 shift hours, cost recovery noise has decayed enough that genuine performance variation is visible. Day 1 tells you almost nothing; Day 3 gives a clear picture.

Last moment to act: The sacrifice decision (add Friday) must be made Wednesday evening. Day 4 gives better data but no time to act within the current week.

Day 3 is where these constraints cross: enough data to project reliably, and the last moment to enact a correction.

9.4 Property 3: Why Adding a Day Beats Chasing Better Trips

Adjusting how many days you work moves CNHR 14× more effectively than improving trip rates. This is a mathematical consequence of large fixed costs, not a data accident.

Why 'Work Smarter' Does Not Apply Here

In many jobs, earning more per hour beats working longer. For PHV driving, the opposite holds. The £220 weekly fixed cost creates leverage: each extra day spreads the same burden across more hours.

Improving p by £1/hr raises CNHR by £0.79. Working one extra day raises it comparably per unit δ . But δ can change by a full day while p fluctuates randomly. The sacrifice rule (adding Friday) is the correct response to underperformance — not surge-chasing.

10. The Monitoring Table

Each week follows this progression:

After shift	Cumul. target	If below	If above
Day 1 (Mon)	£203	Note only — too early to tell	On track
Day 2 (Tue)	£406	Trend forming — be alert	On track
Day 3 (Wed)	£608	FLAG: sacrifice a rest day	Rest days protected
Day 4 (Thu)	£811	Record & assess for next week	NLW compliant

10.1 The Sacrifice Rule

When below £608 at Day 3, convert one rest day to working for that week only:

Priority	Day	Demand Score	Notes
1st choice	Friday	10/20	Highest rest-day demand. Most likely to recover the shortfall.
2nd choice	Saturday	8/20	Moderate demand. Use only if Friday insufficient.
Never	Sunday	6/20	Structurally dead 04:00–08:00. Always protected.

A sacrifice converts the week from 4-on-3-off to 5-on-2-off for that week only. Maximum one sacrifice per week.

Regime change: Two consecutive sacrifice weeks → the assumed p is not achievable. Revert to $d=13$ fallback permanently and reassess.

11. The Fallback Pattern (d=13)

If p consistently falls below £203/day, the schedule reverts to 13 working days per cycle. This is the reason the cycle is 3 weeks long — 13 days cannot be distributed evenly across identical weeks.

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Week 1	WORK	WORK	WORK	WORK	REST	WORK	REST
Week 2	WORK	WORK	WORK	WORK	REST	WORK	REST
Week 3	WORK	WORK	WORK	REST	REST	REST	REST

Structure: Weeks 1–2 are heavier (5 working days each), with rest on Friday and Sunday — preserving weekend bookends. Saturday is an isolated working day. Week 3 is a recovery week (3 days, Mon–Wed), followed by a 4-day rest block Thursday–Sunday.

Why Friday and Sunday? Friday has the highest demand among rest days (10/20), but resting on Friday and Sunday gives two non-consecutive breaks — one at the start and one at the end of the weekend — while capturing Saturday's demand. Sunday is structurally dead in the early morning (04:00–08:00), making it the weakest earning day.

Maximum consecutive working days: 4 (Mon–Thu in weeks 1–2). Same $K=4$ as the primary. Week 3 maximum is 3 (Mon–Wed).

11.1 Fallback Metrics

Metric	Primary (d=12)	Fallback (d=13)
Daily floor	£203	£199
CNHR(e)	£12.25/hr	£12.39/hr
NLW headroom	+£0.04/hr	+£0.18/hr
τ (free hrs per hr worked)	1.55	1.35
π marginal	£14.64/hr	£14.36/hr
Monthly net	£1,844	£2,021
η required	≥ 0.55	≥ 0.59

The fallback earns more per month (£2,021 vs £1,844) but τ drops from 1.55 to 1.35 (less freedom per hour worked) and η rises from 0.55 to 0.59 (more study required per free hour). More financial headroom, less time for study.

When to Use the Fallback

The fallback is not a failure — it is a pre-planned gear change. If daily earnings average £190 instead of £203, the framework shifts to d=13 (NLW-compliant at £199/day). The architecture was designed with this contingency built in.

12. Regulatory and Platform Constraints

The 40-hour week operates within three constraint layers. None are currently binding.

Uber platform: Disconnects after 10 hours of trip time, then 6-hour offline break. At $r = 0.687$, a 10-hour clock shift produces only 6.87 hours of trip time — well under the cap. You would need a 14.5-hour shift to hit it.

Working Time Regulations: 11 hours rest between shifts, 48-hour weekly average (opt-out available), 24 hours rest per 7-day period. The current pattern provides 13 hours between shifts and full rest days every week.

Practical fatigue: Hours 9–10 of a shift carry disproportionate accident risk. The 10-hour shift is a safety boundary, not just a scheduling convenience.

13. What Remains to Be Validated

The framework is trial-ready. The mathematics are verified, the tables are consistent, and the monitoring protocol is operational. Outstanding parameters require real-world data:

Parameter	Current Status	Resolution
ρ (day shift rate)	Target £29.55/hr, not yet measured	2–3 weeks of trial data
α (engagement)	1.26 from night data; day may differ	Re-measure during trial
K=4 (consecutive days)	Untested for day shifts	Observe fatigue over 2–3 weeks
$v_{\text{per_hour}}$	Naive £4.20; needs refinement	Correlate charging receipts with hours
η (study conversion)	Required ≥ 0.55 ; unobserved	Track weekly during trial

The trial validates the inputs; the architecture is sound. If ρ is higher than expected, the framework reduces required days — converting the surprise into free time. If lower, the fallback activates via the sacrifice rule.

14. Document Reference

The framework is a layered set of documents. Information flows one direction: core → applications → evaluation.

Document	Layer	Question It Answers
CORE-001	Core	How do we measure earnings? (CNHR, MNHR, attractor, convergence)
APP-001 (SOM)	Application	How many days, what pattern? (Costs, δ , monitoring, sacrifice rule)
APP-002	Application	Is this schedule liveable? (τ , π , $H(\delta)$, η)
ARCH-001	Architecture	How does it fit together? (Properties 1–3, control system, history)
cnhr_calculator.py	Reference	What are the exact numbers? (Single source of truth)
Work Schedule PDF	Operational	What do I do each day? (One-page printable)
This document	Companion	What does it all mean?

End of Companion Guide.